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United States Patent Application Publication

20250284326

Kind Code

A1

Publication Date

September 11, 2025

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MODULE MOUNTING STRUCTURE AND MINER

Abstract

The present application provides a module mounting structure and a miner. The module mounting structure comprises a first mounting structure and a second mounting structure; the first mounting structure comprises at least one slide rail comprising a connection portion and an extension portion, a sliding groove being formed between the extension portion and the first module; an insertion notch is provided on the extension portion; the second mounting structure comprises at least one hook; at least a portion of the hook may extend through the insertion notch into the sliding groove; when the hook and the insertion notch are staggered in position, the hook located in the sliding groove hooks onto the extension portion. The module mounting structure and the miner according to the present application have advantages such as convenient and simple mounting and the saving of time and efforts.

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Family ID: 94871260

Appl. No.: 19/070578

Filed: March 05, 2025

Foreign Application Priority Data

CN 202420424950.6

Mar. 05, 2024

Publication Classification

Int. Cl.: G06F1/18 (20060101)

U.S. Cl.:

CPC G06F1/181 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of Chinese Patent Application No. 202420424950.6 filed on Mar. 5, 2024, the contents of which are incorporated herein by reference in their entirety.

FIELD

[0002] The present application relates to the field of mounting of structures, and particularly to a module mounting structure and a miner.

BACKGROUND

[0003] Many products available in the market all achieve functions of such products via mutual cooperation between several modules. For example, in the field of miners for earning Bitcoins etc., a miner usually comprises a computing power module, a power supply module etc., wherein the power supply module powers the computing power module. A fixing structure needs to be provided to mount the power supply module and the computing power module together. In the prior art, screws are usually employed to achieve affixation between the modules. The field of miners is no exception. However, the fixing operations with screws are complicated, and the mounting duration is long especially in a case where there are a large number of screws, which do not facilitate improving the mounting efficiency.

SUMMARY

[0004] Based on the present state of the art, a main object of the present application is to provide a module mounting structure and a miner.

[0005] In order to achieve the above object, the present application employs the following technical solutions: the application provides a module mounting structure for achieving the mounting of a first module and a second module to each other, wherein the module mounting structure comprises a first mounting structure disposed on a side surface of the first module and a second mounting structure disposed on a side surface of the second module; the first mounting structure comprises at least one slide rail comprising a connection portion and an extension portion, the extension portion being connected to the first module via the connection portion, a sliding groove being formed between the extension portion and the first module; an insertion notch is provided on the extension portion; the second mounting structure comprises at least one hook at a position corresponding to the slide rail; the hook is connected to the second module; at least a portion of the hook may extend through the insertion notch into the sliding groove; when the hook and the insertion notch are staggered in position, the hook located in the sliding groove hooks onto the extension portion so as to achieve the mounting and fixing of the first module and the second module to each other.

[0006] Preferably, the second module comprises a metal shell; the hook is formed by stamping of part of the metal shell; the hook comprises a body portion and a hook body portion, the body portion is fixedly connected with the second module at both ends of the slide rail in the extension direction; the hook body portion is connected to the body portion in a direction perpendicular to the extension direction of the slide rail; in the extension direction of the slide rail, a size of the hook body portion is smaller than that of the insertion notch.

[0007] Preferably, the slide rail comprises a plurality of slide rails, and a number of the insertion notches disposed on the extension portion of each of the slide rails is one or more; a number of hooks included by the second mounting structure is smaller than or equal to that of the insertion notches; upon mounting, positions of the hooks correspond to those of the insertion notches.

[0008] Preferably, the slide rails comprise a first slide rail and a second slide rail arranged spaced apart, and the insertion notches of the first slide rail and the insertion notches of the second slide rail open towards each other; the hook comprises a body portion and a hook body portion

connected to each other, and the hook body portion is connected to the second module via the body portion; the hook body portions of the hooks corresponding to the first slide rail and the second slide rail face towards each other.

[0009] Preferably, one of the first mounting structure and the second mounting structure further comprises a mounting strip, the other of the first mounting structure and the second mounting structure further comprises a lug, a mounting hole is provided at an end of the mounting strip, and a fixing structure is fixed in the mounting hole through the lug; or the mounting hole mating with the lug is provided at an end of the slide rail.

[0010] Preferably, the slide rails comprise a first slide rail, a second slide rail and a third slide rail which are disposed at intervals and extend in a horizontal direction, the third slide rail being located between the first slide rail and the second slide rail; the first slide rail and the second slide rail are disposed at a top and a bottom of the side surface of the first module, respectively; the first mounting structure is integrally formed on the first module; the second mounting module is integrally formed on the second module.

[0011] The application further provides a miner comprising the module mounting structure and the first module and the second module described above; the first module and the second module are fixedly connected by the module mounting structure.

[0012] Preferably, the miner further comprises a third module fixed on the first module and/or the second module; the first module and/or the second module and/or the third module comprises a box body and a lid body having a rectangular parallelepiped shape, the lid body being used for closing an opening of the box body; the box body comprises a first end panel, a first side panel and a second side panel, the first side panel and the second side panel are connected to opposed ends of the first end panel, and a distance between the two side panels is smaller than a size of the first end plate so that an end of the lid body may abut against the end panel; the side panel comprise a side panel body and a guide structure, the guide structure being fixed at an end of the side panel body adjacent to the first end panel; a distance between the guide structure and a bottom of the box body gradually reduces in a direction away from the first end panel towards the first end panel; the lid body comprises a top plate, a first side plate and a second side plate, the first side plate and the second side plate being connected to opposed ends of the top plate; a fastening strip is disposed at a bottom of an inner wall of each of the first side plate and the second side plate, and a distance between the two fastening strips is equal to that between outer walls of the two side plate bodies; a distance between inner walls of the first side plate and the second side plate is greater than or equal to that between outer walls of the guide structures on the first side panel and the second side panel; a height of the fastening strips is smaller than or equal to a minimum distance between the guide structure and the bottom of the box body.

[0013] Preferably, a length of the guide structure is less than 40% of the length of the side panel; a bottom surface of the guide structure is inclined at an angle in a range of 0° - 45° .

[0014] Preferably, the first module is a computing power module, the second module is a power supply module, and the third module is a control module; a size of the side surface of the first module coincides with that of the side surface of the second module, or the size of the side surface of the second module is smaller than that of the side surface of the first module.

[0015] The present application provides the module mounting structure and the miner; the insertion notches are provided on the slide rail of one module of the module mounting structure; the provision of the insertion notches helps the user to extend the hooks on the other module through the insertion notches into the sliding groove, so that the user needn't move the slide rail in the entire length to complete the fixation of the first module and second module to each other. Particularly in a case where the modules included in the miner have a large weight, the present application saves the user's time and efforts and exhibits convenient and simple mounting.

[0016] Other advantageous effects of the present application will be described by introducing specific technical features and technical solutions in specific embodiments. Those skilled in the art

can understand the advantageous technical effects resulting from the technical features and technical solutions by reading through the introduction of these technical features and technical solutions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] Hereinafter, preferred embodiments of a module mounting structure and a miner according to the present application will be described with reference to the following figures. In the figures:

[0018] FIG. 1 is a perspective structural view of a miner according to the present application.

[0019] FIG. 2 and FIG. 3 are a perspective structural view and a right view of a housing of a first module included in the miner according to the present application.

[0020] FIG. 4 and FIG. 5 are a perspective structural view and a left view of a shell of a second module included in the miner according to the present application.

[0021] FIG. 6 is a structural schematic view of engagement of a sliding groove and a hook included in the module mounting structure according to the present application.

[0022] FIG. 7 is a perspective structural view of a third module of the miner according to the present application.

[0023] FIG. 8 and FIG. 9 are exploded views of a box body and a lid body included in the third module at different angles.

DESCRIPTION OF REFERENCE NUMERALS

[0024] **1**: a miner; **10**: a first module; **11**: a slide rail; **111**: a first slide rail; **112**: a second slide rail; **113**: a third slide rail; **114**: a connection portion; **1141**: a mounting hole; **115**: an extension portion; **1151**: an in insertion notch; **12**: a mounting strip; **20**: a second module; **21**: a hook; **211**: a body portion; **212**: a hook body portion; **30**: a third module; **301**: a module box; **31**: a box body; **311**: a first end panel; **312**: a second end panel; **313**: a first side panel; **3131**: a side panel body; **3132**: a guide structure; **314**: a second side panel; **32**: a lid body; **321**: a top plate; **323**: a first side plate; **3231**: a fastening strip; **324**: a second side plate; **322**, a first end plate; **202**: a lug; **101**: a housing; **201**: a shell.

DETAILED DESCRIPTION OF EMBODIMENTS

[0025] The present application will be described based on embodiments, but the present application is not only limited to these embodiments. In the following detailed depictions of the present application, some specific details are presented in detail. In order to avoid confusing the spirit of the present application, well-known methods, processes, procedures and elements are not described in detail.

[0026] In addition, those having ordinary skill in the art should appreciate that the figures are provided herein for illustrative purposes and are not necessarily drawn to scale.

[0027] Unless the context clearly requires otherwise, words such as “comprise” and “contain” throughout in the whole description and the claim set should be understood as having the meaning of containing rather than exclusive or exhaustive, i.e., as having the meaning of “including but not limited to”.

[0028] In the description of the present application, it is to be understood that the terms “first”, “second” etc. are used for descriptive purposes only and are not to be understood as indicating or implying relative importance. In addition, in the description of the present application, the meaning of “a plurality of” means two or more unless otherwise specified.

[0029] The present application provides a module mounting structure for achieving the mounting of a first module and a second module to each other. The module mounting structure comprises a first mounting structure disposed on a side surface of the first module and a second mounting structure disposed on a side surface of the second module.

[0030] The first mounting structure comprises at least one slide rail comprising a connection portion and an extension portion, the extension portion being connected to the first module via the connection portion, a sliding groove being formed between the extension portion and the first module; an insertion notch is provided on the extension portion.

[0031] The second mounting structure comprises at least one hook at a position corresponding to the slide rail; the hook is connected to the second module.

[0032] At least a portion of the hook may extend through the insertion notch into the sliding groove; when the hook and the insertion notch are staggered in position, the hook located in the sliding groove hooks onto the extension portion so as to achieve the mounting and fixing of the first module and the second module to each other.

[0033] The application further provides a miner comprising the module mounting structure and the first module and the second module described above; the first module and the second module are fixedly connected by the module mounting structure.

[0034] The module mounting structure and the miner according to the present application are described in detail as follows.

[0035] The present application provides a module mounting structure for achieving the mounting of a first module **10** and a second module **20** to each other. The module mounting structure comprises a first mounting structure disposed on a side surface of the first module and a second mounting structure disposed on a side surface of the second module **20**.

[0036] The first mounting structure comprises at least one slide rail **11** comprising a connection portion **114** and an extension portion **115**, the extension portion **115** being connected to the first module **10** via the connection portion **114**, a sliding groove being formed between the extension portion **115** and the first module **10**; an insertion notch **1151** is provided on the extension portion **115**.

[0037] The second mounting structure comprises at least one hook **21** at a position corresponding to the slide rail **11**; the hook **21** is connected to the second module **20**.

[0038] At least a portion of the hook **21** may extend through the insertion notch **1151** into the sliding groove; when the hook **21** and the insertion notch **1151** are staggered in position, the hook **21** located in the sliding groove hooks onto the extension portion **115** so as to achieve the mounting and fixing of the first module **10** and the second module **20** to each other.

[0039] The application further provides a miner **1** comprising the module mounting structure and the first module **10** and the second module **20** described above; the first module **10** and the second module **20** are fixedly connected by the module mounting structure.

[0040] The module mounting structure and the miner **1** according to the present application are described in detail as follows.

[0041] Referring to FIG. **1**, the present application provides a miner **1** for performing data processing. The miner **1** comprises a first module **10**, a second module **20** and a third module **30**, wherein the first module **10** and the second module **20** are disposed side by side and fixedly connected to each other via the module mounting structure; specifically, a side surface of the first module **10** and a side surface of the second module **20** achieve the fixing of the first module **10** and the second module **20** via the module mounting structure. The third module **30** is fixed on top of the first module **10** and the second module **20**.

[0042] As one embodiment, the first module **10** is a computing power module, the second module **20** is a power supply module, and the third module **30** is a control module. It may be appreciated that the functions of the first module **10**, the second module **20** and the third module **30** are not limited; the first module **10** may also be a power module or a control module or other functional module; the second module **20** may also be a computing power module or a control module or other functional module; and the third module **30** may be a computing power module, a power supply module or other functional module.

[0043] It may be appreciated that the module mounting structure between the first module **10** and

the second module **20** may be adapted for fixed connection of the side surfaces of any two modules. The first module **10** and the second module **20** are not limited to the modules included by the miner **1**, but may also be modules included by any other product.

[0044] Referring to FIG. 2 and FIG. 3, the first module **10** comprises a housing **101** in which elements are received. The module mounting structure comprises a first mounting structure disposed on a side surface of the housing **101** adjacent to the second module **20**. The first mounting structure comprises at least one slide rail; in the present embodiment, the first mounting structure comprises a first slide rail **111**, a second slide rail **112** and a third slide rail **113** which are disposed at intervals and extend in a horizontal direction, wherein the first slide rail **111** and the second slide rail **112** are disposed at the top and the bottom of the side surface of the first module **10**, respectively, and the third slide rail **113** is disposed between the first slide rail **111** and the second slide rail **112**.

[0045] The first slide rail **111**, the second slide rail **112** and the third slide rail **113** all comprise a connection portion **114** and an extension portion **115**, the extension portion **115** is connected to a side surface of the first module **10** via the connection portion **114**, a sliding groove is formed between the extension portion **115** and the first module **10**, and at least one insertion notch **1151** is disposed on the extension portion **115**. The number of the insertion notches **1151** on the first slide rail **111**, the second slide rail **112** and the third slide rail **113** may stay consistent or inconsistent, as long as it is ensured that each slide rail comprises at least one insertion notch **1151**. As shown in FIG. 2 and FIG. 3, the first slide rail **111** and the second slide rail **112** are each provided with two insertion notches **1151**, and the third slide rail **113** is provided with one insertion notch **1151**. Preferably, the insertion notches **1151** on the first slide rail **111**, the second slide rail **112** and the third slide rail **113** are aligned in a vertical direction.

[0046] It may be appreciated that the number of slide rails may be adjusted according to factors such as the size and weight of the products. In some embodiments, the number of slide rails may be one, two or more.

[0047] In some embodiments, the insertion notches **1151** of at least two slide rails open towards each other, i.e., the opening directions of the sliding grooves are opposed. As a specific embodiment, the insertion notches **1151** of the first slide rail **111** and the second slide rail **112** open towards each other so that the relative movement between the first module **10** and the second module **20** may be restricted in the vertical direction.

[0048] As one embodiment, the first mounting structure further comprises a mounting strip **12** extending in a horizontal direction and fixed to the side surface of the housing **101**. A mounting hole **121** is disposed on an end face of the mounting strip **12** and used for a fixed connection between the first module **10** and the second module **20**.

[0049] As a variant embodiment, the mounting hole **1151** is disposed at an end face of the slide rail **11**. As another variant embodiment, the end face of the slide rail **11** is provided with the mounting hole **1151**, and the end face of the mounting strip **12** is provided with the mounting hole **121**.

[0050] As one embodiment, the housing **101** is made of a metal, and the first mounting structure is integrally formed with the housing **101**.

[0051] Referring to FIG. 4 and FIG. 5, a second mounting structure is provided on a side surface of the second module **20**, and the second mounting structure comprises at least one hook **21** at a position corresponding to the slide rail **11**; the hook **21** is connected to the second module **20**.

[0052] Referring to FIG. 6, at least a portion of the hook **21** may extend thorough the insertion notch **1151** into the sliding groove; when the hook **21** and the insertion notch **1151** are staggered in position, the hook **21** located in the sliding groove hooks onto the extension portion **115** so as to achieve the mounting and fixing of the first module **10** and the second module **20** to each other. It may be appreciated that under the action of gravity, the portion of the hook **21** located in the sliding groove hooks onto the extension portion **115**.

[0053] As one embodiment, the insertion notches **1151** on the second slide rail **112** at the bottom of

the side surface of the first module **10** open upwards, so that the second module **20** may hook onto the extension portion **115** by the hook **21** under the action of gravity.

[0054] As one embodiment, the hook **21** comprises a body portion **211** and a hook body portion **212**, the hook body portion **212** is connected to the second module **20** via the body portion **211**, and the body portion **211** is fixedly connected with the second module **20** at both ends of the slide rail **11** in the extension direction; the hook body portion **212** is connected to the body portion **211** in a direction perpendicular to the extension direction of the slide rail **11**.

[0055] As one embodiment, the second module **20** comprises a metal shell **201**; the hook **21** is formed by stamping of part of the metal shell.

[0056] As one embodiment, the number of hooks **21** included by the second mounting structure is smaller than or equal to that of the insertion notches **1151**; upon mounting, the positions of the hooks **21** correspond to those of the insertion notches **1151**, so that all the hooks **21** can extend through the insertion notches **1151** into the sliding groove. After the hooks **21** extend through the insertion notch **1151** into the sliding groove, the first module **10** or the second module **20** is moved so that the positions of the hooks **21** are offset from those of the insertion notches **1151**, and the hooks **21** hook onto the extension portion **115** under the action of gravity.

[0057] It may be appreciated that in the extension direction of the slide rail **11**, the size of the hook body portion **212** is necessarily smaller than that of the insertion notch **1151** so that the hook body portion **212** may extend through the insertion notch **1151** into the sliding groove.

[0058] As one embodiment, when the insertion notches **1151** on different slide rails **11** open towards each other, the hook body portions **212** corresponding to positions of the different slide rails **11** open towards each other so that the hook body portions **212** can simultaneously enter the sliding grooves through the corresponding insertion notches **1151**.

[0059] As one embodiment, the second mounting structure further comprises a lug **202** through which a fixing structure is fixed in the mounting hole **121** or the mounting hole **1151**, thereby fixing the first module **10** and the second module **20** in position in the extension direction of the slide rail **11** and preventing relative movement therebetween. It may be appreciated that the position of the mounting strip **12** provided with the mounting hole **121** and the position of the lug **202** is interchangeable, for example, the mounting strip **12** is disposed on the side surface of the second module **20**, and the lug **202** is disposed on the first module **10**. The lug **202** may be located at a front end or a rear end of the second module.

[0060] Referring to FIG. 7, FIG. 8 and FIG. 9, the third module **30** comprises a box body **31** and a lid body **32** having a rectangular parallelepiped shape, the lid body **32** being used for closing an opening of the box body **31**.

[0061] As one embodiment, the box body **31** comprises a first end panel **311**, a second end panel **312**, a first side panel **313** and a second side panel **314**, wherein the first end panel **311**, the first side panel **313**, the second end panel **312** and the second side panel **314** are connected to form a rectangular parallelepiped. The first end panel **311** and the second end panel **312** are located at one pair of opposed ends of the third module **30**, and the first side panel **313** and second side panel **314** are located at the other pair of opposed ends of the third module **30**.

[0062] As one embodiment, the first side panel **313** and second side panel **314** are connected to opposed ends of the first end panel **311**, with a distance between the two side panels being smaller than the size of first end panel **311** so that the end of lid body **32** may abut against the end plate **322**. In use, one end of the lid body **32** covers the box body **31** from an end close to the second end panel **312**; the lid body **32** is pushed so that the lid body **32** moves from the end close to the second end panel **312** to a position close to the first end panel **311**. Since the distance between the two side panels is smaller than the size of the first end panel **311**, when the lid body **32** is pushed to the position of the first end panel **311**, the end of the lid body **32** will abut against the first end panel **311**, thereby limiting a movement stroke of the lid body **32**.

[0063] As one embodiment, the side panel comprises a side panel body **3131** and a guide structure

3132. The guide structure **3132** is fixed to an end of the side panel body **3131** adjacent to the first end panel **311**, and a distance between the guide structure **3132** and the bottom of the box body **31** gradually reduces in a direction away from the first end panel **311** towards the first end panel **311**. That is, the bottom surface of the guide structure **3132** has a ramp structure. As one embodiment, the bottom surface of the guide structure **3132** is inclined at an angle in a range of 20° - 45° .

[0064] As one embodiment, the lid body **32** comprises a top plate **321**, a first side plate **323**, a second side plate **324** and a first end plate **322**. The top plate **321** has a rectangular shape. The first side plate **323**, the second side plate **324** and the first end plate **322** are connected to the top plate **321** in three different directions, respectively. It may be appreciated that in a case where the second end panel **312** is present, the first end plate **312** may be omitted; in a case where the first end plate **322** is present, the second end panel **312** may be omitted.

[0065] As one embodiment, the first side plate **323** and the second side plate **324** are connected to opposed ends of the top plate **321**, a fastening strip **3231** is disposed at the bottom of an inner wall of each of the first side plate **323** and second side plate **324**, and a distance between the two fastening strips **3231** is equal to that between outer walls of the two side plate bodies. When the lid body **32** covers the box body **31**, the fastening strips **3231** contact the side plate bodies so that the lid body **32** tightly covers the box body **31**. A height of the fastening strips **3231** is smaller than or equal to a minimum distance between the guide structure **3132** and the bottom of the box body **31**. Upon mounting, the fastening strip **3231** and the guide structure **3132** do not interfere with each other without causing a mounting obstacle. It may be appreciated that the height of the fastening strip **3231** here refers to the height of the fastening strip **3231** at a position corresponding to the guide structure **3132**, and the heights at remaining positions is not limited herein.

[0066] As one embodiment, the fastening strip **3231** may be rectangular, or may have other regular or irregular shape, such as triangular, trapezoidal, etc.

[0067] As one embodiment, a distance between inner walls of the first side plate **323** and the second side plate **324** is greater than or equal to that between outer walls of the guide structures **3132** on the first side panel **313** and the second side panel **314**, so that the lid body **32** can smoothly cover the box body **31**.

[0068] As one embodiment, a length of the guide structure **3132** is less than 40% of the length of the side plate; a bottom surface of the guide structure **3132** is inclined at an angle in a range of 0° - 45° .

[0069] As one embodiment, the third module **30** may be secured on the top of the first module **10** or the second module **20**.

[0070] As one embodiment, the first module **10** and/or the second module **20** may also employ the structure of the box body **31** and the lid body **32** as described above.

[0071] As one embodiment, sizes of the side surfaces of the first module **10** and the second module **20** coincide with each other, or the size of the side surface of the second module **20** is smaller than that of the side surface of the first module **10**.

[0072] As one embodiment, after the lid body **32** covers the box body **31**, the first end plate **322** and the second end panel **312** are locked with screws to achieve the locking of the lid body **32** and the box body **31**.

[0073] When the lid body **32** needs to cover the box body **31**, the user may hold, with a hand, one end of the lid body **32** where the first end plate **322** lies, and places the other end opposite to the first end plate **322** vertically downward from a position of the side plate adjacent to the guide structure **3132** so that the other end opposite to the first end plate **322** covers the box body **31**; the user continues to apply a force to the end where the first end plate **322** lies to move the lid body **32** towards the first end panel **311**, the inclined bottom surface of the guide structure **3132** may cooperate with the fastening strips **3231** to guide the lid body **32** to slide. When the lid body **32** abuts against the first end panel **311**, the lid body **32** duly covers the box body **31**.

[0074] In a state that the lid body **32** completely, duly covers the box body **31**, when the lid body

32 needs to be removed, since the guide structure **3132** blocks the fastening strip **3231** in the vertical direction, the lid body **32** cannot be removed directly in the vertical direction; the user needs to pull the lid body **32** in a direction away from the first end panel **311**; as an operation manner, the user may lift one end of the lid body **32** vertically at the position of the second end panel **312**; since the bottom surface of the guide structure **3132** is an inclined structure, when the inclination angle of the fastening strip **3231** is less than that of the bottom surface of the guide structure **3132**, the guide structure **3132** will not cause blockage to the fastening strip **3231**, which helps the user to pull the lid body **32** in the direction away from the first end panel **311**; after the lid body **32** moves a distance (i.e., when there is no guide structure **3132** above the fastening strip **3231**), the user may directly removes the lid body **32**.

[0075] Those skilled in the art can appreciate that the above-described preferred solutions may be freely combined and superimposed without conflict.

[0076] It should be appreciated that the above-described embodiments are merely exemplary rather than non-limiting, and that various obvious or equivalent modifications to and substitutes for the above-described details made by those skilled in the art without departing from the basic principles of the present application, are all included within the scope of the appended claims of the present application.

Claims

1. A module mounting structure for achieving mounting of a first module and a second module to each other, wherein the module mounting structure comprises a first mounting structure disposed on a side surface of the first module and a second mounting structure disposed on a side surface of the second module; the first mounting structure comprises at least one slide rail comprising a connection portion and an extension portion, the extension portion being connected to the first module via the connection portion, a sliding groove being formed between the extension portion and the first module; an insertion notch is provided on the extension portion; the second mounting structure comprises at least one hook at a position corresponding to the slide rail; the hook is connected to the second module; at least a portion of the hook may extend through the insertion notch into the sliding groove; when the hook and the insertion notch are staggered in position, the hook located in the sliding groove hooks onto the extension portion so as to achieve the mounting and fixing of the first module and the second module to each other.
2. The module mounting structure according to claim 1, wherein the second module comprises a metal shell; the hook is formed by stamping of part of the metal shell; the hook comprises a body portion and a hook body portion, the body portion is fixedly connected with the second module at both ends of the slide rail in the extension direction; the hook body portion is connected to the body portion in a direction perpendicular to the extension direction of the slide rail; in the extension direction of the slide rail, a size of the hook body portion is smaller than that of the insertion notch.
3. The module mounting structure according to claim 1, wherein the slide rail comprises a plurality of slide rails, and a number of the insertion notches disposed on the extension portion of each of the slide rails is one or more; a number of hooks included by the second mounting structure is smaller than or equal to that of the insertion notches; upon mounting, positions of the hooks correspond to those of the insertion notches.
4. The module mounting structure according to claim 3, wherein the slide rails comprise a first slide rail and a second slide rail arranged spaced apart, and the insertion notches of the first slide rail and the insertion notches of the second slide rail open towards each other; the hook comprises a body portion and a hook body portion connected to each other, and the hook body portion is connected to the second module via the body portion; the hook body portions of the hooks corresponding to the first slide rail and the second slide rail face towards each other.
5. The module mounting structure according to claim 1, wherein one of the first mounting structure

- and the second mounting structure further comprises a mounting strip, the other of the first mounting structure and the second mounting structure further comprises a lug, a mounting hole is provided at an end of the mounting strip, and a fixing structure is fixed in the mounting hole through the lug; or the mounting hole mating with the lug is provided at an end of the slide rail.
- 6.** The module mounting structure according to claim 1, wherein the slide rails comprise a first slide rail, a second slide rail and a third slide rail which are disposed at intervals and extend in a horizontal direction, the third slide rail being located between the first slide rail and the second slide rail; the first slide rail and the second slide rail are disposed at a top and a bottom of the side surface of the first module, respectively; the first mounting structure is integrally formed on the first module; the second mounting module is integrally formed on the second module.
- 7.** A miner, wherein the miner comprises the module mounting structure and the first module and the second module according to claim 1; the first module and the second module are fixedly connected by the module mounting structure.
- 8.** The miner according to claim 7, wherein the miner further comprises a third module fixed on the first module and/or the second module; the first module and/or the second module and/or the third module comprises a box body and a lid body having a rectangular parallelepiped shape, the lid body being used for closing an opening of the box body; the box body comprises a first end panel, a first side panel and a second side panel, the first side panel and the second side panel are connected to opposed ends of the first end panel, and a distance between the two side panels is smaller than a size of the first end plate so that an end of the lid body may abut against the end panel; the side panel comprise a side panel body and a guide structure, the guide structure being fixed at an end of the side panel body adjacent to the first end panel; a distance between the guide structure and a bottom of the box body gradually reduces in a direction away from the first end panel towards the first end panel; the lid body comprises a top plate, a first side plate and a second side plate, the first side plate and the second side plate being connected to opposed ends of the top plate; a fastening strip is disposed at a bottom of an inner wall of each of the first side plate and the second side plate, and a distance between the two fastening strips is equal to that between outer walls of the two side plate bodies; a distance between inner walls of the first side plate and the second side plate is greater than or equal to that between outer walls of the guide structures on the first side panel and the second side panel; a height of the fastening strips is smaller than or equal to a minimum distance between the guide structure and the bottom of the box body.
- 9.** The miner according to claim 8, wherein a length of the guide structure is less than 40% of the length of the side panel; a bottom surface of the guide structure is inclined at an angle in a range of 0° - 45° .
- 10.** The miner according to claim 8, wherein the first module is a computing power module, the second module is a power supply module, and the third module is a control module; a size of the side surface of the first module coincides with that of the side surface of the second module, or the size of the side surface of the second module is smaller than that of the side surface of the first module.
- 11.** The miner according to claim 9, wherein the first module is a computing power module, the second module is a power supply module, and the third module is a control module; a size of the side surface of the first module coincides with that of the side surface of the second module, or the size of the side surface of the second module is smaller than that of the side surface of the first module.
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